Cardiac Deformation Lab

Image-Based Cardiac Deformation Mapping & Kinematic Strain Analysis

Analysis ID:	356aa5c4-6ec1-48c0-96a3-a019e9ace2e6	**Analysis Type:**	Single-Pair Differential Strain Analysis
Execution Date:	October 15, 2026 at 17:27:41 UTC	**Motion Model:**	Dense Farnebäck Optical Flow
Spatial Calibration:	Not provided; displacement values reported in image pixels (px)	**Calibration Status:**	Image-Space Only
Analysis Region:	Region of Interest (Polygon) • 1891 px	**Region Scope:**	ROI Analysis (Polygon)
Reference Frame (ED):	test_roi_f1.png	**Target Frame (ES):**	test_roi_f2.png

1. Displacement & Kinematic Motion

Metric Parameter	Value (Pixel Units)	Description
Mean Displacement	1.5978 px	Average Euclidean magnitude of displacement vector field
Maximum Displacement	7.0085 px	Peak regional apparent motion amplitude across the field
Minimum Displacement	0.0000 px	Lowest regional motion magnitude observed
Displacement Std. Dev.	2.0160 px	Spatial heterogeneity of tissue displacement

2. Small-Deformation Strain Tensor & Principal Fields

Strain Component	Mean	Min	Max	Std Dev
Normal Exx (du/dx)	9.5186e-11	N/A	3.2209	N/A
Normal Eyy (dv/dy)	0.0000	N/A	2.7782	N/A
Shear Exy (0.5*(du/dy+dv/dx))	5.2982e-10	N/A	1.8148	N/A
Principal Strain 1 (Major)	0.0868	N/A	N/A	N/A
Principal Strain 2 (Minor)	-0.0868	N/A	N/A	N/A

3. Advanced Deformation Metrics & Geometric Projection

Metric Parameter	Observed Value	Physical / Mathematical Interpretation
Motion Coherence	0.8715	Cosine consistency of motion vectors within local 5x5 spatial kernels
Strain Gradient Magnitude	0.0771	Spatial rate of change of principal strain fields
Strain Anisotropy Index	0.6129	Directional preference of tissue deformation (|e1 - e2| / (|e1| + |e2|))

Strain Entropy	2.5914	Shannon entropy describing dispersion of regional strain histograms
Deformation Energy	0.0917	Pseudo-elastic strain energy density surrogate (0.5 * Tr(eps * eps))
Radial Deformation (Mean)	-1.3885	Motion projection along radial axes from ventricular centroid approximation
Circumferential Deformation (Mean)	1.1848e-07	Motion projection orthogonal to radial vectors (tangential shortening)

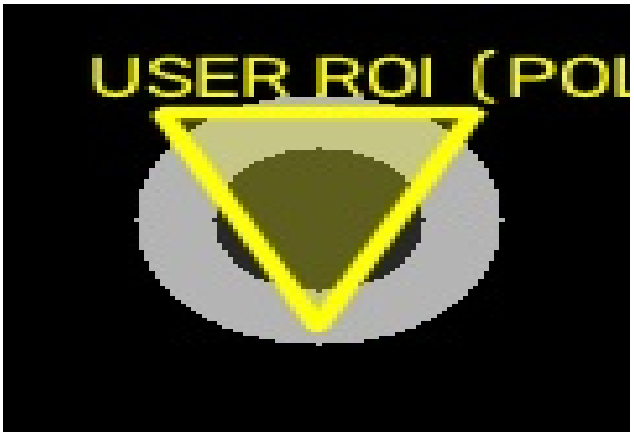
5. Cardiac Deformation Fingerprint (10-Dimensional)

Dimension Symbol	Canonical Name	Value
D_mean	Mean Displacement	N/A
D_max	Maximum Displacement	N/A
epsilon_mean	Representative Mean Strain	N/A
epsilon_max	Representative Max Strain	N/A
epsilon_principal	Representative Principal Strain	N/A
AI	Strain Anisotropy Index	N/A
H	Strain Entropy	N/A
G_epsilon	Strain Gradient Magnitude	N/A
C	Motion Coherence	N/A
TSI	Temporal Stability Index	N/A

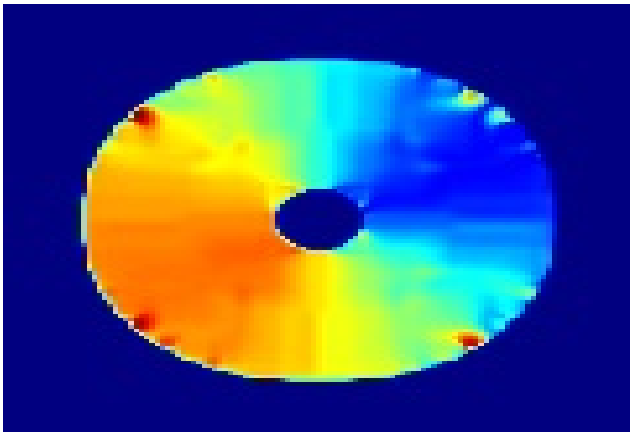
6. Spatial Deformation Maps & Motion Field Visualizations

Two-dimensional colormaps generated by OpenCV and matplotlib engines illustrating spatial gradients and vector trajectories.

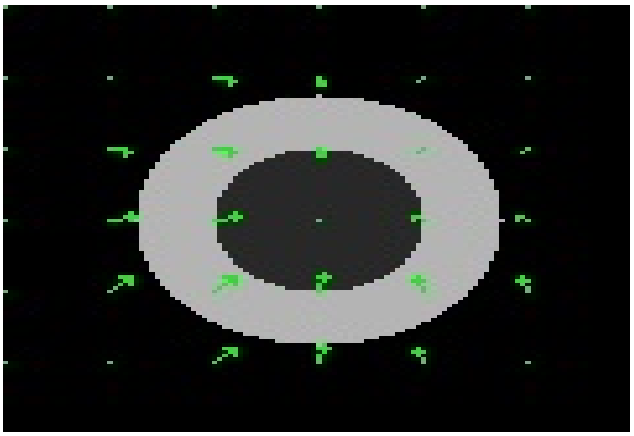
ROI Overlay Boundary



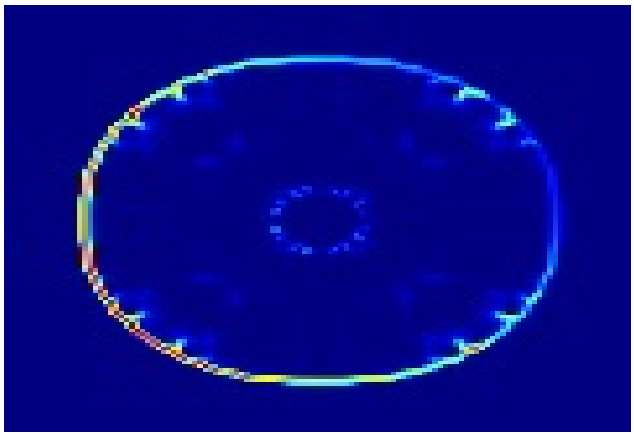
Displacement Heatmap



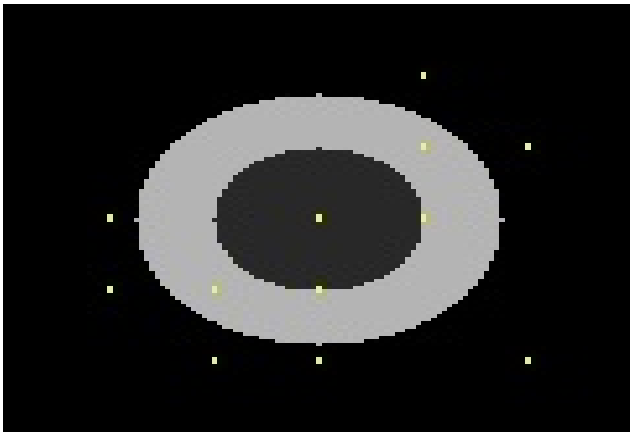
Motion Vector Map



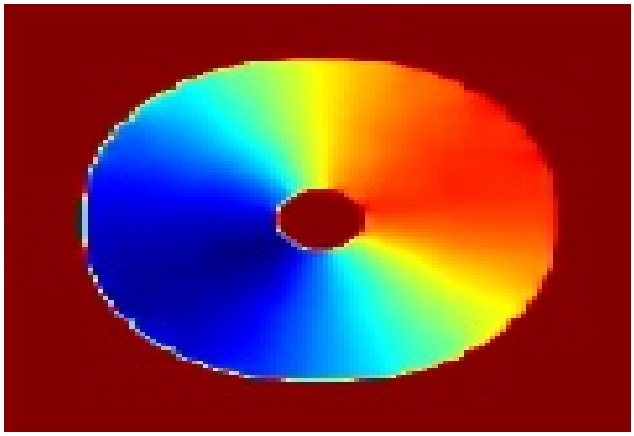
Principal Strain Heatmap



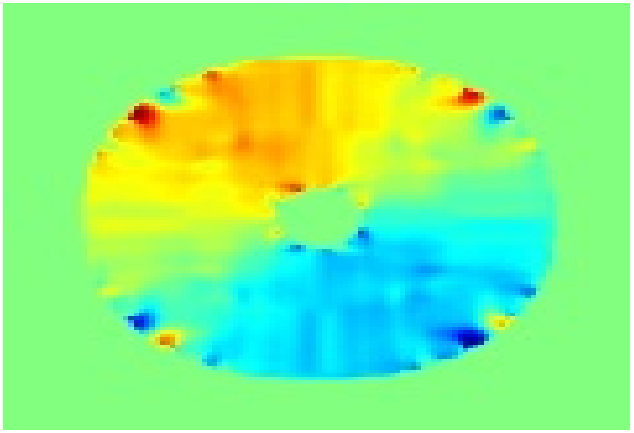
Principal Direction Map



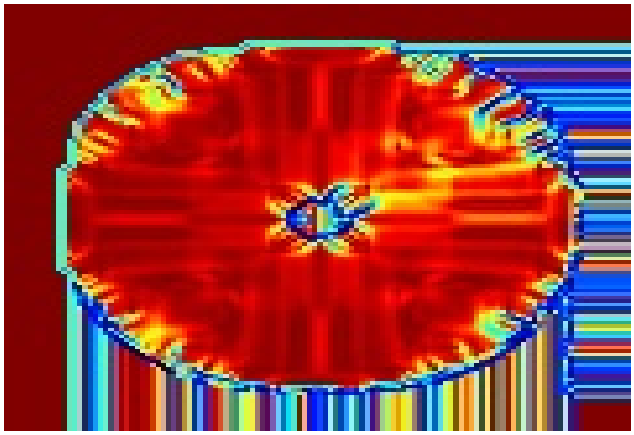
Radial Deformation Map



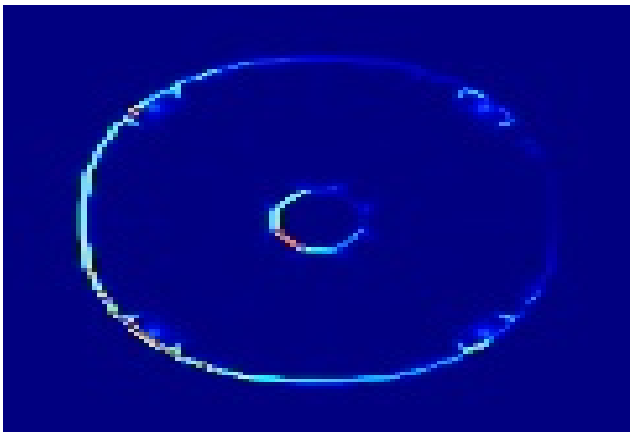
Circumferential Deformation



Motion Coherence Map



Deformation Energy Map



7. Technical Interpretation & Research Prototype Disclaimer

Methodological Scope: The metrics, vector fields, and spatial colormaps detailed in this report represent apparent image-space deformation computed via Farnebäck dense optical flow and classical continuum strain tensor decomposition. Measurements are expressed in image coordinate units (pixels) and relative mathematical differentials.

RESEARCH ONLY — NO CLINICAL DIAGNOSTIC ASSERTIONS:

This software is an experimental computational research prototype. All values, including the Cardiac Deformation Fingerprint (10-D) and Temporal Stability Index (TSI), are algorithmic descriptors of pixel kinematics. This system does NOT provide medical diagnoses, clinical risk assessments, normal/abnormal classifications, or treatment recommendations. This report must NOT be used as a medical device or diagnostic decision-making instrument.